

Climate-Linked Bonds (CLBs):

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What this paper aims to do

Big idea: Use *financial engineering*—sovereign **climate-linked bonds (CLBs)**—to help solve three problems at once.

- ▶ **Information:** let prices reveal expectations/risk premia about climate outcomes.
- ▶ **Incentives:** tie sovereign cash flows to climate variables so policy improvements cut expected payouts $\Rightarrow$ cheaper financing now.
- ▶ **Insurance:** give exposed investors a hedge against slow-moving climate damages.

Design sketch: nominal bond + linear climate *swap* leg; payoffs indexed to a measurable climate variable (global or local temperature, emissions), with optional coupon floors.

What is a ClimateLinked Bond (CLB)?

Definition

A sovereign (or supranational) bond whose **cash flows adjust mechanically** with a **verifiable climate index** (e.g., temperature anomaly, own country GHG emissions path, water/sealevel).
No use of proceeds restriction (unlike green bonds).

Simple explanation: CLB = Nominal bond + climate swap

Structure & payoff

$$\text{Coupon}_t = c_0 + b_1 \Delta I_t \quad \text{with optional floor/cap,}$$

ΔI_t is the change in the climate index

c_0 baseline coupon with no change in the index

b_1 sensitivity to the climate index

$$\text{Price} = \mathbb{E}[\text{cash flows}] - \text{climate risk premium (covariance with SDF)}.$$

Main Ideas

CLBs deliver **price discovery**, **policy alignment**, and **risk sharing**.

Approach of the paper

- ▶ Policy alignment and incentives are not modeled
- ▶ The price discovery is shown with a simple SDF exposition
- ▶ The potential for risk sharing is shown in an equilibrium model
- ▶ The possible magnitude of the market is estimated based on another model, which is inspired by the equilibrium model
- ▶ Numerical estimates suggest CLB could be 3 % of GDP for many countries

Main Ideas

My summary take

- ▶ This is ambitious and important
- ▶ **Information** revealed depends on the exact financial claim...and the potential market for that financial claim
- ▶ **Incentives** are weak if the index is global; stronger with controllable indices (own emissions)
- ▶ **Insurance** faces first-order *basis risk*; current markets suggest demand is concentrated among directly exposed agents.

Comment: Economics of the 3% of GDP Claim

Model micro side (equilibrium bond price where supply and demand are equal)

Hedging demand (“Eq. 17”):

$$f_d = \delta \frac{d_1}{b_1}$$

Exposed share δ wants to hedge damages $d_1 \Delta T$ with a payoff $b_1 \Delta T$.

Investment/supply (“Eq. 18”):

$$f_s = \frac{R_f B^T - (b_0 + b_1 \mu)}{\alpha \sigma^2 b_1^2}$$

Supply depends on expected return vs. risk (σ^2) and risk aversion (α).

Comment: Economics of the 3% of GDP Claim

- ▶ assume that the expected temperature increases over a projection horizon of h are μ
- ▶ one-period expected losses as a percentage of GDP are then given by $\frac{d_1\mu}{h}$
- ▶ total issued amount of the bonds in local currency, as a percentage of GDP, is f
- ▶ coupon adjustment mechanism such that the expected coupon payment per period is $\frac{b_1\mu}{h}$
- ▶ each period, a fraction $1/m$ of the climate-linked bonds will redeem back, where m is the average duration of the bonds outstanding

Back-of-the-envelope potential market size (“Eq. 26”): coverage identity

$$\frac{d_1\mu}{h} = \left(\frac{b_1\mu}{h}\right)f + \left(\frac{1}{m}\right)f \Rightarrow f = \frac{d_1\mu/h}{b_1\mu/h + 1/m}$$

Matches *expected climate losses per year* to *expected CLB flows* (coupon + redemptions).

Comment: Economics of the 3% of GDP Claim

- ▶ **Different objects:** (17)–(18) are *market-clearing quantities and price*; (26) is a *budget coverage target*.
- ▶ **Units:** (17)–(18) in portfolio/wealth units; (26) sets f as % of GDP (stock-to-flow accounting).
- ▶ **What “covers” losses?** (26) counts *principal amortization* $\frac{1}{m}f$; **insurance economics suggests matching only climate-contingent coupons**. Including the redemption each period conflates financing with insurance

Coupons-only coverage (drop f/m)

$$\frac{d_1 \mu}{h} = \left(\frac{b_1 \mu}{h} \right) f \quad \Rightarrow \quad \boxed{f = \frac{d_1}{b_1}}$$

- ▶ Aligns with the micro *hedge ratio*. Under baseline $b_1 \approx d_1$, the target is $f \approx 100\%$ of GDP, not a few percent.

Comment: What's the market failure?

Why do CLBs not already exist? Who uses existing climate-linked derivatives?

Temperature (Heating/Cooling Degree Days) futures & options (CME)

Hedgers: utilities, district heating, energy marketers, weathersensitive retail, airports/events.

Liquidity: banks' weather desks, market makers, commodity/macro HFs.

OTC parametric weather (temp/rain/wind/solar)

Hedgers: wind/solar power producers, hydro, ag/agribusiness, construction, ski/leisure, logistics/mining.

Providers: reinsurers, bank structuring desks.

Catlinked (Industry loss warranty/Insurance linked security)

Buyers: insurers/reinsurers, sovereign/pool facilities.

Sellers: ILS funds, pensions, multistrategy HFs.

Carbon allowance derivatives

Compliance: power generators, industrials, airlines/shipping, fuel suppliers.

Financial: banks, commodity merchants, HFs, AM/ETFs, prop firms.

Weathershaped power/profile products

Hedgers: renewable owners, utilities/retailers, corporate Power Purchase Agreements, storage.

Liquidity: power traders/market makers.

Precipitation/riverflow/waterlevel indices

Hedgers: hydro generators, barging/shipping, water utilities, ag coops.

Providers: reinsurers/ILS, banks.

Reality: usage skews to direct P&L exposure and allowance markets work only with credible political enforcement

Comment: What's the market failure?

Why do CLBs not already exist?

- ▶ Empirical reality (who trades weather risk) suggests narrower natural holders than assumed
- ▶ Weather/parametric markets are used mostly by directly exposed entities. Limited evidence for demand for “temperature beta” or “emission beta”
- ▶ **Basis risk:** Damages are heterogeneous, non-linear, and location/sector specific; a national temperature or emissions index is an imperfect hedge for many holders.
- ▶ I would like to see more focus on how the government can provide a unique role.

Comment: What's the market failure?

Solution and a potential problem...

- ▶ **Make index controllable:** Link to own emissions or hedge variables relevant for investors
- ▶ This provides a credible benefit once a stock of CLBs exists. Better policy today reduces the *PV of climate coupons* on outstanding CLBs $\Rightarrow$ immediate drop in the market value of debt $\Rightarrow$ in the limit - the government should insure more risks to have "skin in the game"?
- ▶ Government is effectively the issuer of the bond with an embedded short swap position that pays out based on a climate-linked variable $\Rightarrow$ government provides insurance $\Rightarrow$ **government formalizes its role as insurer of last resort...** (for the incentive to work they must not reinsure that risk....)

Minor Comments

- ▶ **Information:** Identifying beliefs vs. risk premia from prices is under-identified without strong assumptions...
- ▶ **Incentives:** Global indices yield weak policy control; free-riding remains unless the index is local/controllable.
- ▶ **Moral Hazard:** The authors recognize positive feedback - if governments issue these bonds then they have an incentive to reduce payouts. If people can insure risk using the government, and they don't bear all the consequences... will they change their behavior?
- ▶ Why bundle the swap (insurance) with the bond (financing)? The government can also issue nominal bonds and also provide insurance (collecting a premium if investors demand the hedge).

Conclusion

- ▶ This paper lays out clearly the financial economics of climate-linked bonds using our best tools (SDF, equilibrium model)
- ▶ I'll be following this idea with great interest
- ▶ Good luck with the paper